MUHAMAD ARIEF LIMAN

Software Engineer • Backend Specialist

Jakarta, Indonesia • Open to On-site & Hybrid Opportunities

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PROFESSIONAL SUMMARY

Results-driven **Software Engineer** specializing in enterprise backend architecture, distributed financial transaction processing, and scalable microservices utilizing **Java (Spring Boot)**, **NestJS**, and **TypeScript**. Experienced in designing resilient payment systems (Xendit), high-concurrency event platforms (Netflix's Squid Game: The Experience Jakarta), and full-lifecycle membership and transaction engines (FIFGROUPCARD at PT Nostra Solusi Teknologi). Strong background in automated testing with **Testcontainers**, JUnit 5, and CI/CD pipelines, supported by a Master of Computer Science degree (GPA 3.59/4.00) and published international peer-reviewed papers in Deep Learning and Computer Vision.

TECHNICAL SKILLS

Backend & Languages:	**Java (Spring Boot)**, **TypeScript**, **Node.js (NestJS, Express)**, Python, C/C++, PHP, SQL
Testing & Quality:	**Testcontainers**, JUnit 5, Mockito, Automated Integration Testing, Test-Driven Development (TDD)
Databases & Caching:	PostgreSQL, MySQL, Redis, Microsoft SQL Server, Query Optimization & Schema Design
DevOps & Cloud:	**Docker**, Docker Compose, **CI/CD (GitHub Actions)**, Linux/Ubuntu Server, Git Version Control
Integrations & Architecture:	**Xendit Payment Gateway API**, RESTful API Design, Microservices, Webhook Handling, CMS Backends
Frontend & Mobile:	Vue.js, React.js, JavaScript (ES6+), HTML5/CSS3, Bootstrap, Android, Flutter

PROFESSIONAL EXPERIENCE

Software Engineer • **PT Nostra Solusi Teknologi** (Project: FIFGROUPCARD / FIF) — 2024 – Present | Jakarta, Indonesia

- Engineered end-to-end financial transaction processing workflows and business validation logic for the **FIFGROUPCARD** enterprise platform.
- Architected and implemented centralized administrative management systems covering **Members**, **Store Partnerships**, **Product Catalogs**, and dynamic **Promo/Discount** engines.
- Maintained high-availability membership infrastructure and data consistency, supporting seamless member verification, loyalty rewards, and core transaction services.
- Collaborated in an Agile enterprise environment delivering scalable backend services with clean architecture and strict quality standards.

Backend Engineer & Solutions Architect • **Production & Commercial Platforms** — 2023 – Present | Jakarta, Indonesia

- **Netflix's Squid Game: The Experience Jakarta (Agora Mall Thamrin Nine):** Architected high-performance landing page APIs and administrative CMS for the official portal (squidgameexperienceid.com), handling high concurrency for contestant ticketing and session scheduling.
- **Asha Ship (Marine Ticketing & CMS):** Built comprehensive vessel booking, scheduling, and passenger manifest management platform using **NestJS**, **TypeScript**, and **PostgreSQL**, fully integrated with **Xendit Payment Gateway** for automated payments and webhook reconciliations.
- **BasketinAja Mini League V2:** Designed and deployed automated basketball tournament management system for 5-team round-robin formats with real-time match stats recording, automated leaderboard calculations, and MVP metrics using **Vue.js**, **PostgreSQL 15**, and **Docker**.

Web Developer • **Telkom Indonesia** — 2022 – 2023 | Jakarta, Indonesia

- Engineered and maintained internal operational web applications and digital service portals supporting enterprise telecommunications workflows.
- Built reliable backend integrations and responsive user interface components adhering to corporate security and design standards.

EDUCATION

Master of Computer Science (S2) • Bina Nusantara University (BINUS) — Fast-Track Program • **GPA: 3.59 / 4.00** — 2022 – 2024

Bachelor of Computer Science (S1) • Bina Nusantara University (BINUS) — Undergraduate Degree in Computer Science — 2019 – 2023

ACADEMIC RESEARCH & PUBLICATIONS (GOOGLE SCHOLAR)

1. Handwritten Character Recognition Using Vision Transformer and Convolutional Neural Networks — *International Journal of Advanced Science, Engineering and Information Technology (JOIV), Vol. 14, 2024.*

2. Face Mask Detection Using Deep Convolutional Neural Networks — *Elsevier Procedia Computer Science, Vol. 227, pp. 642-649, 2023.*

3. Performance Comparison of Classical Machine Learning and Deep Learning for Emotion Recognition — *Revue d'Intelligence Artificielle (RIA), Vol. 38, No. 2, 2024.*